

Junaid Ahmed Mohammed

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with **5+ years** of experience building scalable **Azure**, **Snowflake**, and **Databricks** lakehouse platforms for analytics and AI/ML workloads. Built high-throughput **ETL/ELT** pipelines processing **12+ TB** of enterprise retail data across **625+ stores** in 60+ countries using **PySpark**, **Kafka**, **Airflow**, **Structured Streaming**, and **Delta Lake**. Graduate research focused on distributed streaming systems and **LLM** training pipelines.

TECHNICAL SKILLS

Cloud & Data Platforms : Azure Databricks, Snowflake (Zero-Copy Cloning, Time Travel), ADF, ADLS Gen2, Azure Blob Storage, Azure ML

Data Engineering & Processing : PySpark, Spark SQL, Structured Streaming, SQL Tuning, Apache Kafka, Airflow, Delta Lake, ETL/ELT, Lakehouse Architecture, Medallion Architecture, Data Modeling, Dimensional Modeling, SCD Type 1/2, Data Profiling, Source-to-Target Mapping, Data Governance, Data Lineage, Unit Testing, Regression Testing, CI/CD

Data Formats : JSON, Parquet, ORC, Avro, XML, CSV

Programming & Query Languages : Python, SQL, Scala, T-SQL, PL/SQL

Databases & Warehousing : PostgreSQL, SQL Server, Oracle, Teradata

AI/ML Enablement : Vector Search, pgvector, OpenAI Embeddings, QLoRA Fine-Tuning, LLM Data Pipelines

Tools & Collaboration : Git, GitHub, Azure DevOps, Confluence, Azure Key Vault, Power BI

PROFESSIONAL EXPERIENCE

NISUM TECHNOLOGIES, INC.

Nov 2018 – Dec 2023

Senior Data Engineer

Hyderabad, India

Supported enterprise retail data platform modernization initiatives for Williams-Sonoma and Macy's across Azure, Snowflake, Databricks, and enterprise ETL/ELT ecosystems supporting analytics, supply chain, fulfillment, and operational reporting workloads.

Client: Williams-Sonoma, Inc. | Enterprise Retail Analytics, Supply Chain, and Fulfillment Platforms | 625+ stores, 60+ countries

- Architected **Azure** and **Snowflake** lakehouse platforms supporting retail analytics, supply chain operations, and customer delivery forecasting across **625+ stores** while processing **12+ TB** of enterprise retail data.
- Deployed and expanded **160+ ADF** and **Databricks** ingestion workflows within Bronze-Silver-Gold **Delta Lake** layers supporting enterprise analytics and operational reporting.
- Refactored **PySpark** transformations to reduce pipeline runtime by **40%** through partition pruning, Z-Ordering, broadcast joins, and Spark SQL tuning.
- Developed **Snowflake** and **Databricks** ingestion pipelines processing **12+ TB** of enterprise retail data across batch and near-real-time workloads.
- Integrated **Oracle**, **Teradata**, **Kafka**, **NetSuite ERP**, and **SFTP** datasets into **ADLS Gen2** through full and incremental load strategies.
- Modeled dimensional warehouse schemas and **SCD Type 2** pipelines using **Snowflake** clustering strategies, reducing infrastructure costs by **25%**.
- Computed carrier and routing dimensions for the **ECDD** customer delivery platform, reducing customer delivery inaccuracies by **30%**.
- Coded **ADF** orchestration pipelines, **Databricks** staging layers, and **Scala-based Snowflake** loaders supporting **Oracle-to-Snowflake WMOS** migration.
- Defined **Snowflake** and **Databricks** data contracts across **Orders**, **Products**, **Stores**, and **Vendor** domains by translating stakeholder and analyst requirements into standardized ingestion workflows.
- Automated **Git-based CI/CD** workflows through **Azure DevOps** integrations while coordinating deployment activities and managing **Azure Key Vault** credentials.
- Prepared curated **Azure ML** feature datasets and downstream data pipelines supporting enterprise analytics and model-training workflows.

- Documented Databricks, Snowflake, and ADF operational runbooks while mentoring junior engineers on onboarding, troubleshooting, and production support activities.
- Resolved SLA-critical ETL failures, production incidents, and data inconsistencies by coordinating with enterprise SRE teams during Azure and Snowflake operational escalations.

Client: Macy's, Inc. | *Early-career data engineer supporting enterprise retail data warehouse*

- Monitored **25+ ADF** pipelines across daily batch windows to maintain enterprise SLA compliance across production ETL workloads.
- Reconciled source-to-target datasets to detect data loss and duplication across Oracle, SQL Server, and downstream warehouse pipelines.
- Debugged pipeline failures and coordinated production fixes with senior engineers during operational escalations and batch-processing incidents.
- Validated pipeline transformations through unit and regression testing prior to enterprise production deployments.
- Profiled **20+ operational tables** in Oracle and SQL Server to certify data readiness across enterprise reporting and ingestion workflows.
- Patched historical reload issues across warehouse pipelines by troubleshooting transformation inconsistencies and failed batch-processing jobs.

INDIANA UNIVERSITY INDIANAPOLIS

Jan 2024 – Dec 2025

Graduate Research Assistant, COMET Lab

Indianapolis, IN

- Engineered an end-to-end streaming ingestion platform using **Kafka** and **PySpark Structured Streaming**, validating **50K+ rows per batch cycle** before loading governed datasets into **PostgreSQL**.
- Scheduled automated model-training workflows through **Airflow** DAGs with threshold-based deduplication logic and containerized orchestration using **Docker Compose**.
- Shipped distributed Spark-submit jobs for revenue and unit-sales forecasting across scalable batch-processing workloads.
- Constructed **LLM** training data pipelines and evaluation infrastructure for **QLoRA** fine-tuning of LLaMA 3.2 and Qwen 2.5 models.
- Benchmarked structured prompt datasets and correctness harnesses validating domain-specific Python code generation across **LLM** evaluation workloads.
- Conducted weekly Python, SQL, and machine learning lab sessions for **60+ students**, reinforcing applied data engineering and machine learning concepts.

PERSONAL PROJECTS

JOB AI MATCHER | *Semantic Job Matching and AI-Enabled Application Platform*

Jan 2026 – Present

Production Beta

- Architected and deployed semantic job-matching APIs using **FastAPI** and **PostgreSQL**, executing vector similarity search through **pgvector** and **OpenAI embeddings**.
- Ranked semantic resume-job matches using **GPT-4o-mini** workflows, generating contextual application guidance and interview preparation responses.
- Designed backend **FastAPI** APIs supporting resume parsing, embedding generation, job deduplication, application tracking, and profile management.

EDUCATION

INDIANA UNIVERSITY

Indianapolis, Indiana

M.S. in Applied Data Science | GPA: 3.97 / 4.0

Jan 2024 – Dec 2025

Coursework: Statistical Learning, Cloud Computing, NLP, Computer Vision, Database Design

CERTIFICATIONS

Snowflake SnowPro Core | Azure Data Engineer Associate (DP-203) | Generative AI 101, Indiana University